Predicting Peak Electricity Demand in Massachusetts

Machine Learning Analysis for Grid Reliability and Capacity Planning

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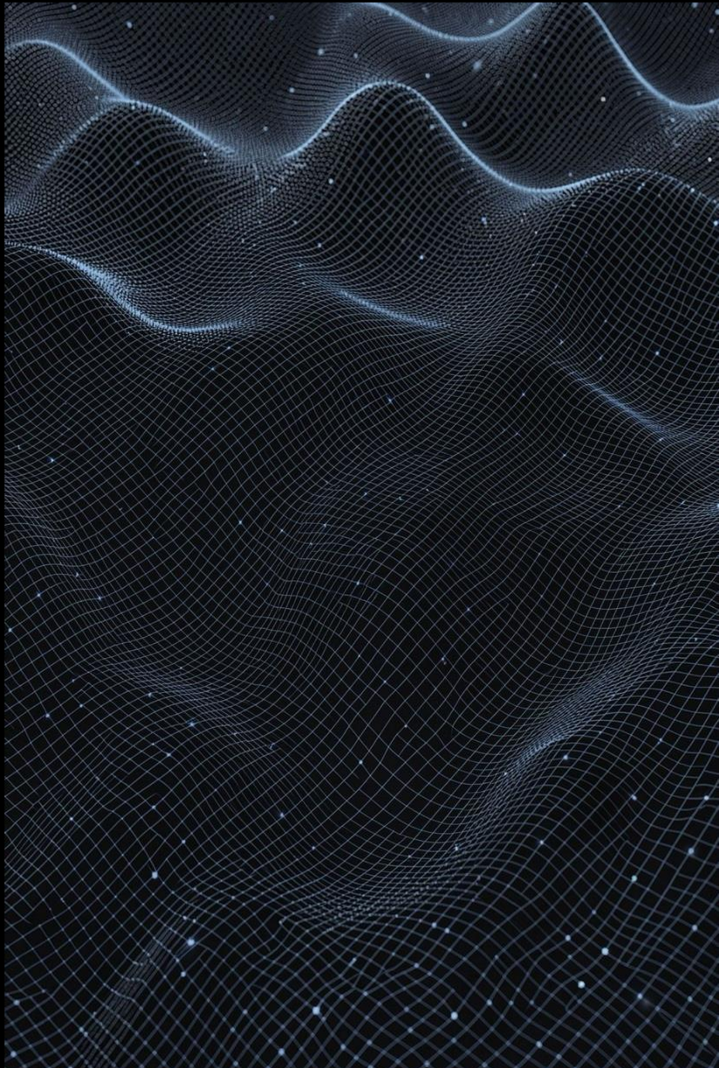


Agenda

01 **Background & Problem**
Why peak demand forecasting matters

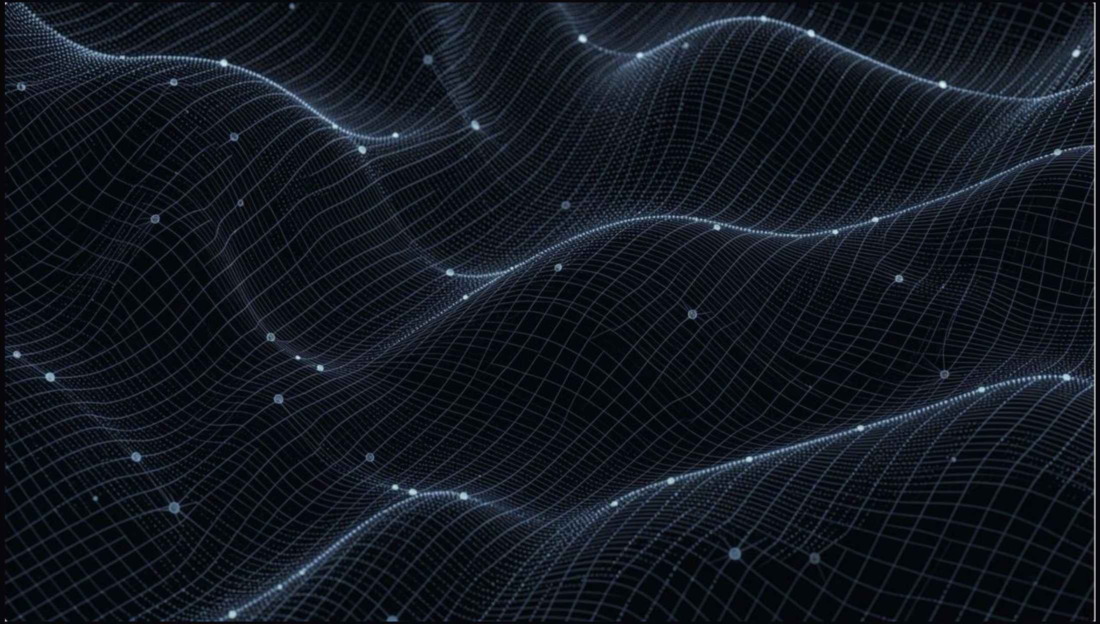
02 **Data Sourcing & Validation**
Real ISO-NE load, weather, renewables

03 **Feature Engineering**
Lags, degree days, calendar proxies

04 **Model Comparison**
Baseline → SARIMAX → ML → Ensemble

05 **Extreme Peak Drivers**
Winter heating vs summer cooling

06 **Limitations & Takeaways**
Caveats and key conclusions



STRATEGIC CONTEXT

Why Peak Demand Matters

The hours of highest demand are not just operational events — they are the defining moments for grid infrastructure decisions.

- **Grid stress peaks** determine what generation capacity must always be available, even if used only a handful of hours per year
- **Renewable variability** adds complexity: solar and wind are weather-dependent, making demand-supply matching harder to predict
- **Accurate forecasting** is the cornerstone of reliable, cost-effective grid planning for years ahead

The Grid Reliability Challenge

Real-Time Balancing

ISO New England must match supply and demand every single hour — there is no margin for error.

Under-Forecasting Risk

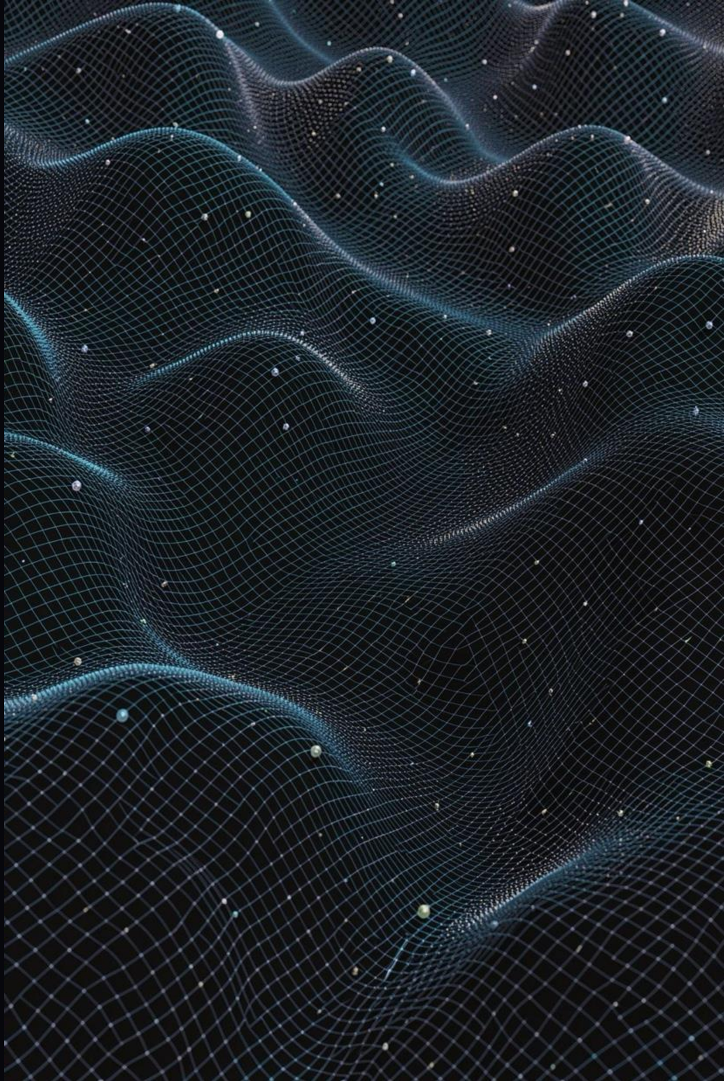
Blackouts and emergency power purchases at extreme spot prices threaten both reliability and budgets.

Over-Forecasting Cost

Excess capacity commitments waste ratepayer dollars and erode trust in planning accuracy.

Infrastructure Implications

Peak demand hours set the benchmark for capacity market pricing and long-term infrastructure investment.



Project Objectives

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Forecast Hourly Demand

Combine weather, load history, renewable generation, and behavioral patterns into a unified prediction model

Identify Peak Triggers

Pinpoint the specific conditions most strongly associated with extreme peak demand events

Seasonal Comparison

Distinguish between winter heating peaks and summer cooling peaks to inform season-specific planning

Data Sources: Real, Public, Validated

Electricity Demand

ISO-NE hourly load data pulled directly from the EIA API — authoritative, granular, and current.

Weather Data

Boston hourly temperature, humidity, wind speed, and solar radiation sourced from Open-Meteo.

Renewable Generation

ISO-NE solar and wind generation figures via EIA API, capturing the variable supply side of the grid.

Consumer Behavior

Calendar features including hour of day, day of week, and weekend/holiday flags to capture behavioral rhythms.

Data Validation: A Critical Discovery

Not all publicly available data is what it claims to be — and this project proved it.

⚠ Initial Kaggle dataset failed validation: synthetic data with flat patterns and zero autocorrelation. It would have invalidated every model result.

The team replaced it entirely with real ISO-NE demand, verified Boston weather records, and actual renewable generation figures. The real data exhibits the strong autocorrelation and seasonal structure that genuine electricity systems produce — and that forecasting models require.

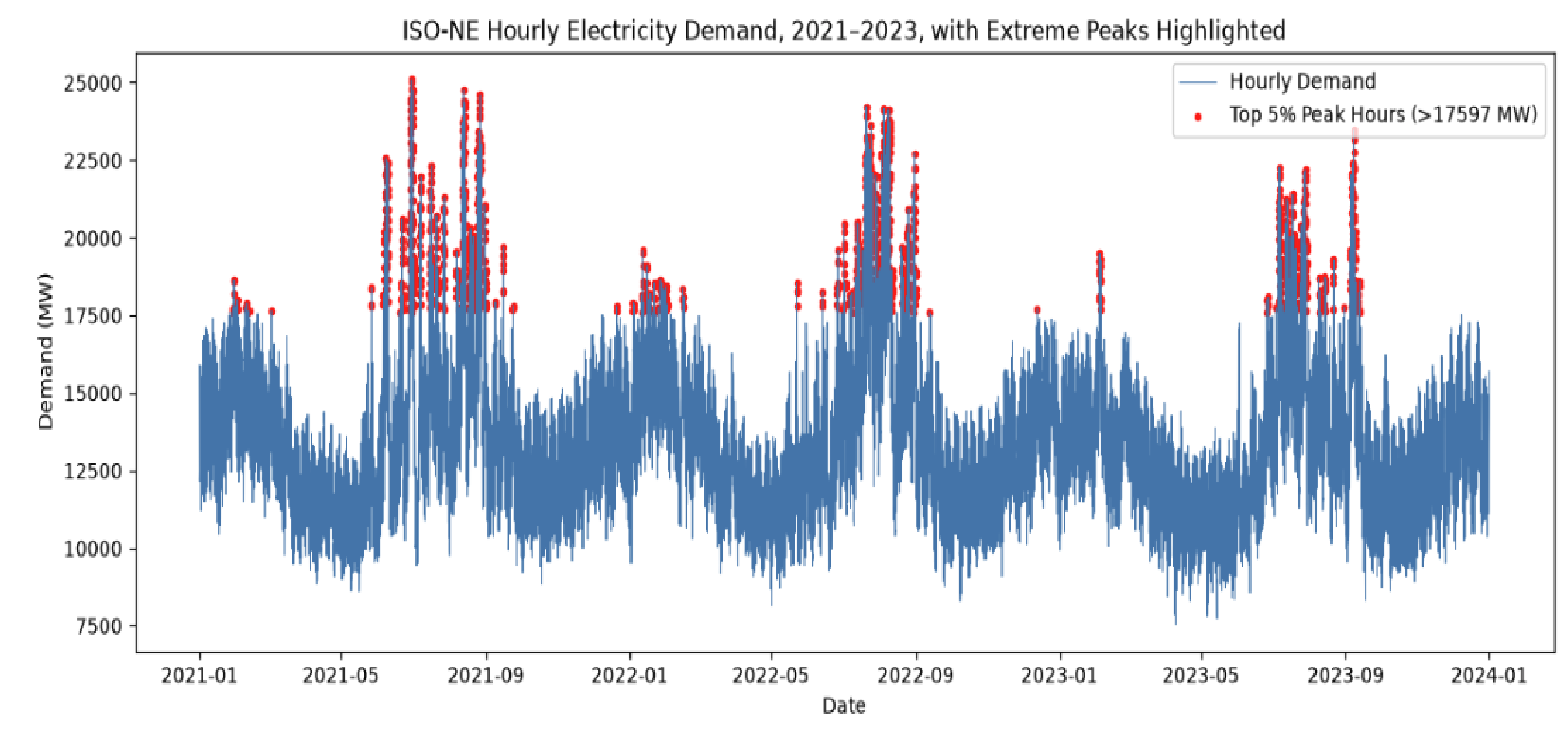
❌ Rejected

Kaggle synthetic dataset — flat patterns, zero autocorrelation

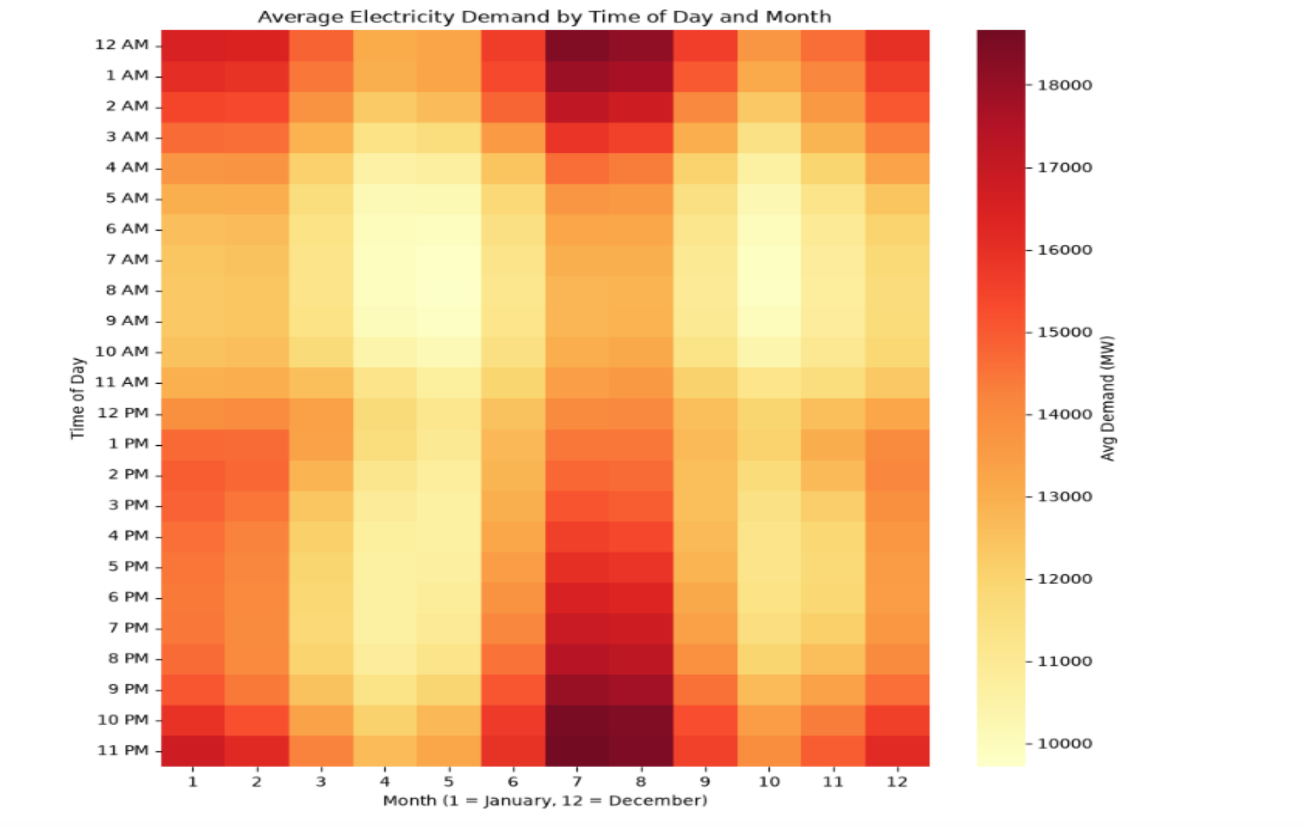
✅ Adopted

Real ISO-NE + Open-Meteo — strong seasonal structure confirmed

ISO-NE Hourly Demand with Extreme Peaks



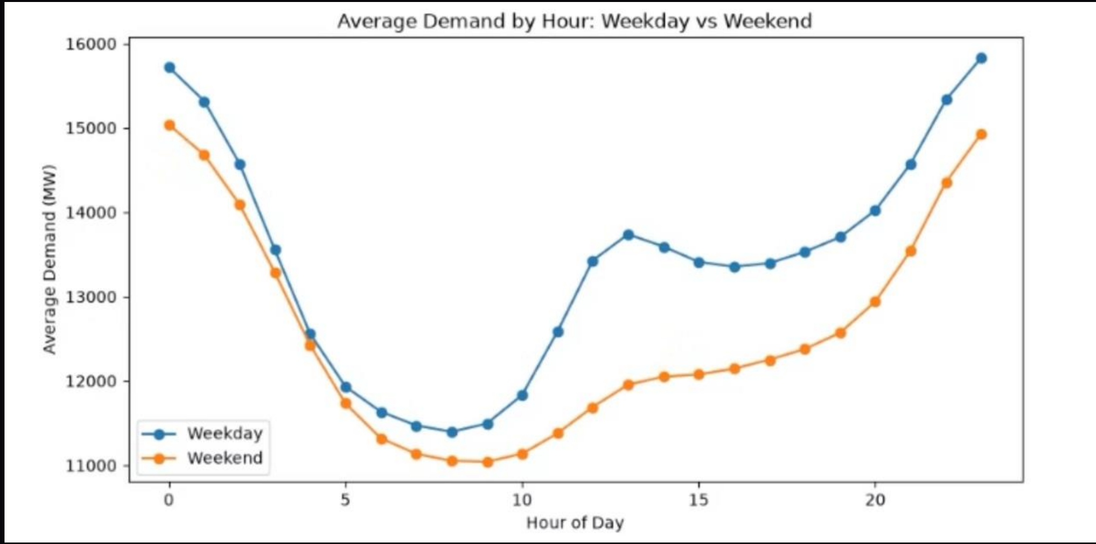
Average Demand by Time of Day & Month



Weekday vs. Weekend Demand

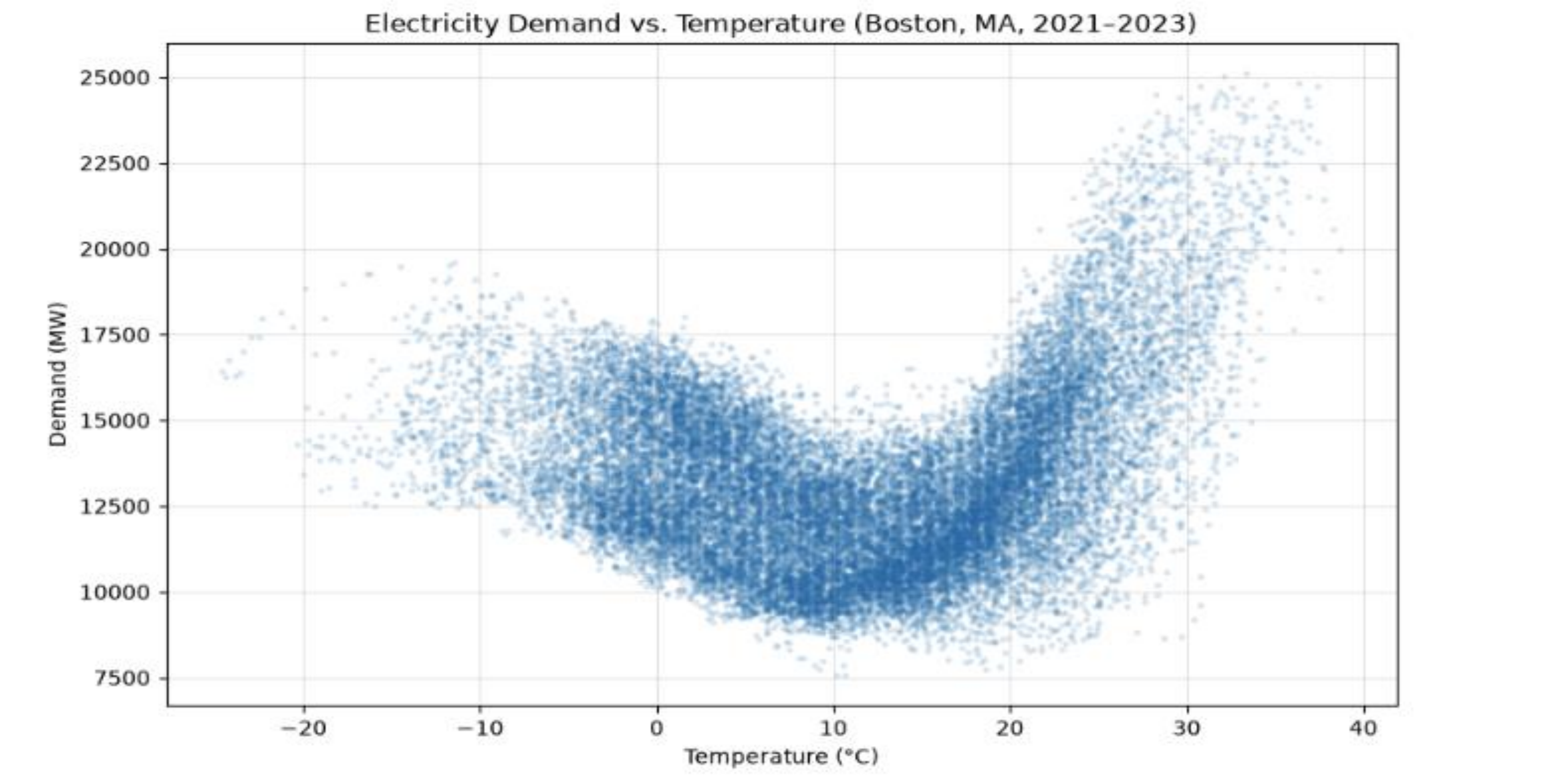
Behavioral patterns leave a measurable imprint on grid load. Weekday commercial and industrial activity drives a distinctly different demand profile compared to weekend residential use.

- Weekday peaks are sharper and earlier, driven by commercial load
- Weekend demand is flatter and shifted later in the day
- Holiday flags capture anomalous low-demand days that would otherwise distort model training



These behavioral features are among the strongest non-weather predictors in the ensemble model.

Demand vs. Temperature (Boston, 2021–2023)



Feature Engineering

Lag Features

1-hour lag

24-hour lag

168-hour (weekly)

Rolling Stats

Rolling averages

Rolling std. dev.

Recent trends

Weather-Derived

Heating degree days

Cooling degree days

Temp interactions

Calendar / Behavior

Hour of day

Day of week

Weekend & holiday

Model Comparison Results

Model	MAE (MW)	RMSE (MW)	MAPE
Baseline (seasonal avg)	1,217.0	1,468.5	10.47%
SARIMAX (90-day)	5,036.2	5,381.1	43.61%
SARIMAX (24-hour)	1,963.5	2,189.2	18.29%
XGBoost	151.2	196.0	1.25%
LightGBM	153.5	199.2	1.27%
Ensemble	**149.1**	**193.7**	**1.23%**

Ensemble: ~8.5× improvement over seasonal-average baseline

Key Modeling Insights

Ensemble Wins

Best MAPE of 1.23%. XGBoost and LightGBM nearly identical; combining them edged out either alone.

SARIMAX Struggled

Underperformed the naive baseline at both 24h and 90-day horizons due to compounding forecast drift.

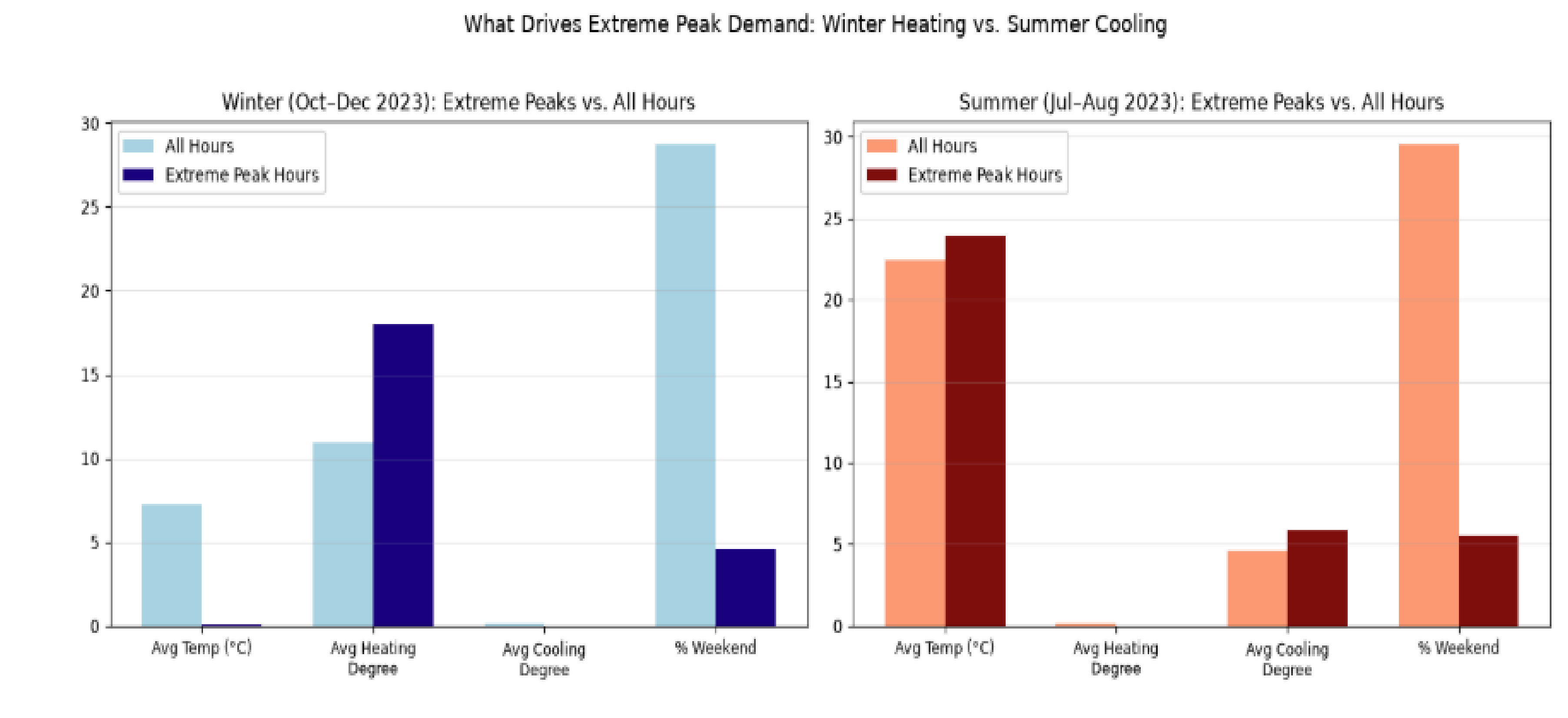
Renewables Limited Impact

Solar and wind contributed <1% combined feature importance — effect largely captured via weather and time.

Net Load Excluded

Demand-minus-renewables feature engineered but excluded to avoid target leakage and inflated performance.

What Drives Extreme Peak Demand?



Seasonal Drivers of Extreme Peaks

Winter (Oct–Dec 2023)

Avg Temp on Peaks
0.6°C vs 7.3°C all hours

Heating Degree
17.4 vs 10.9 (~60% higher)

Timing
Median hour ≈ 7 PM

Weekday Dominance
95.4% of peaks on weekdays

Summer (Jul–Aug 2023)

Avg Temp on Peaks
27.9°C vs 22.4°C all hours

Cooling Degree
9.9 vs 4.6 (>2× higher)

Timing
Median hour ≈ 7 PM

Weekday Dominance
94.5% of peaks on weekdays

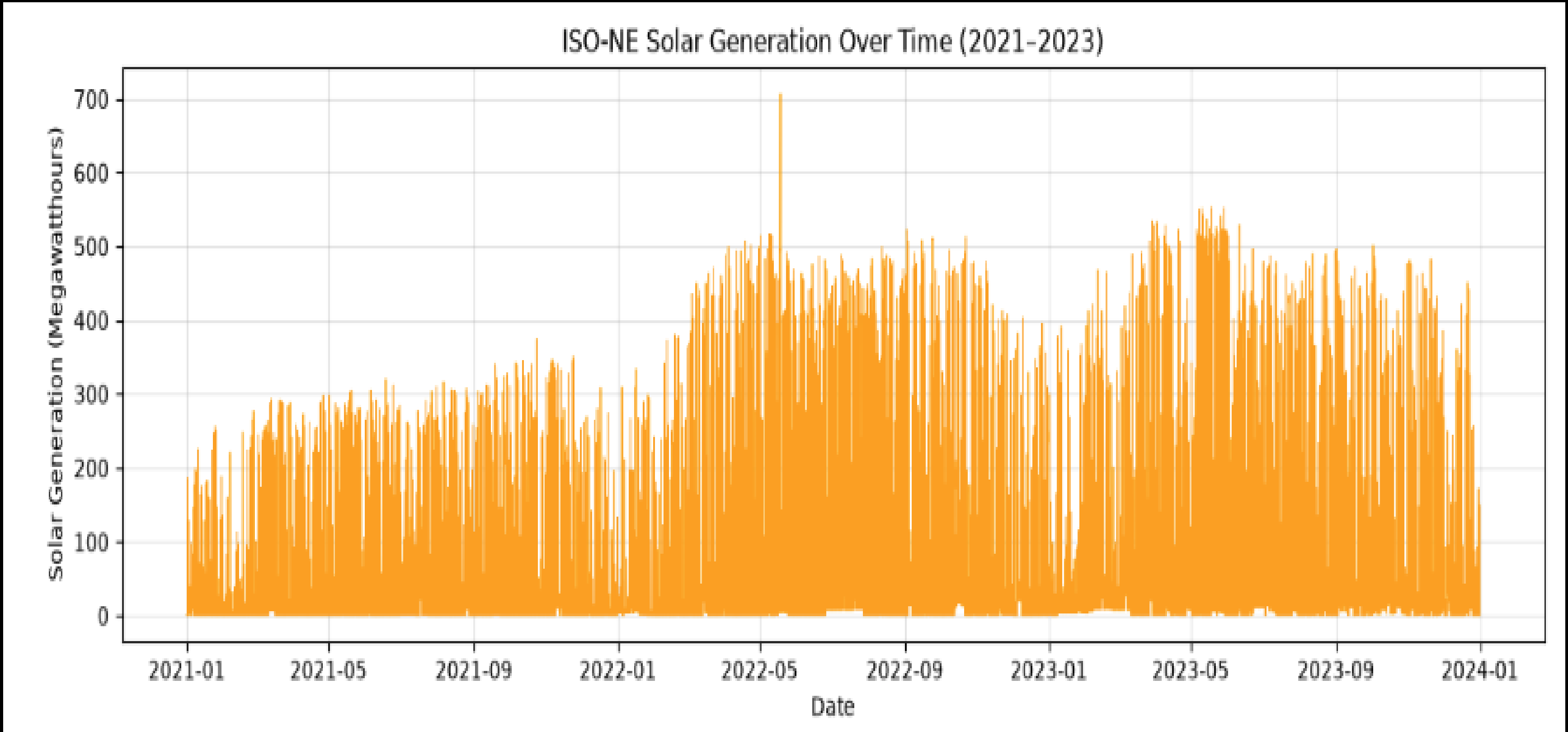
The Common Thread Across Seasons

Early-evening, weekday-dominant signature

Regardless of whether the underlying driver is heating load in winter or cooling load in summer, extreme peaks consistently cluster around 7 PM on weekdays (~95% of cases).

The timing of residential and commercial activity, layered on top of whatever weather extreme a given day presents, is the common thread underlying grid stress events across seasons.

ISO-NE Solar Generation Over Time



Model Performance: 8.5× Better Than Baseline

1.23%	151MW	10.47%
Ensemble MAPE	XGBoost MAE	Baseline MAPE
Best overall performer — combining XGBoost and LightGBM predictions	1.25% MAPE — operationally precise for capacity planning decisions	Seasonal naive benchmark — the model is 8.5× more accurate

What This Means for Operators

At Massachusetts scale, a 1.23% MAPE translates to forecasting errors well under 200 MW on typical days — a precision level that directly reduces reserve margin requirements and emergency procurement costs.

XGBoost

1.25% MAPE · 151.2 MW MAE

LightGBM

1.27% MAPE · 153.5 MW MAE

Seasonal Baseline

10.47% MAPE · reference only

Ensemble accuracy supports confident capacity commitment and reduced reliance on costly emergency reserves.

Limitations

Geographic Proxy

Weather-load relationship established for Boston as a proxy for the broader Massachusetts / ISO-NE region.

SARIMAX Tuning

Limited tuning to fit compute time; a full seasonal SARIMAX may perform better with more resources.

Seasonal Windows

Two 2-month seasonal windows analyzed; a full-year rolling analysis would further strengthen findings.

Feature Comparison

SARIMAX received narrower exogenous inputs than ML models by design — not fully apples-to-apples.

Key Takeaways

1. **High-accuracy forecasting is achievable**
   Ensemble MAPE of 1.23% — an 8.5× gain over seasonal averages — using load lags, weather, and calendar features.

2. **Peaks are structural, not purely weather**
   Extreme demand clusters at ~7 PM on weekdays in both seasons; activity timing amplifies weather stress.

3. **Opposite physical drivers, same pattern**
   Winter peaks = heating load; summer peaks = cooling load. The behavioral overlay is the common thread.

4. **Data integrity first**
   Synthetic data would have produced misleading results. Real public APIs enabled credible, actionable insights.

Thank You

Q&A